# Avaota A1 – BLZ-enabled ZHA & Zigbee2MQTT

Board: Avaota A1 (Allwinner T527 / arm64)

Before starting, flash your ThirdReality Zigbee 3.0 USB Dongle with the latest firmware.

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## Part 1 – Flash Armbian Image

### Option A: Pre-built image (recommended)

Pre-built images include Armbian, USB serial drivers, Docker, Home Assistant, and AI assistant (OpenClaw or PicoClaw). Download from the release page:

| Image | RAM | Includes |
|---|---|---|
| `*_openclaw.img.xz` | 4GB+ | HA + OpenClaw AI |
| `*_picoclaw.img.xz` | 1GB+ | HA + PicoClaw AI |

Steps:

1. Download the image matching your board's RAM.
2. Flash to SD card using balenaEtcher.
3. Insert SD card into Avaota A1, power on.
4. Default credentials: see release notes.

Skip to **Part 2** after boot.

### Option B: Build Armbian from source

For custom configurations or when pre-built images don't meet your needs.

```
# On an x86_64 Linux host with Docker
git clone https://github.com/fangzheli/build.git \
  armbian-build
cd armbian-build

# WSL2 only: enable arm64 emulation
docker run --privileged --rm \
  tonistiigi/binfmt --install arm64

# Build (legacy branch only for avaota-a1)
./compile.sh build \
  BOARD=avaota-a1 BRANCH=legacy \
  RELEASE=noble BUILD_DESKTOP=yes \
  DESKTOP_ENVIRONMENT=xfce \
  KERNEL_CONFIGURE=yes
```

In kernel menuconfig, enable USB serial drivers:

```
Device Drivers --->
  [*] USB support --->
    <M> USB Serial Converter support
    <M> USB CH341 single port serial driver
    <M> USB CP210x family UART Bridge
```

Flash the output image from `output/images/Armbian-unofficial_*.img`.

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## Part 2 – Verify Dongle Detection

Plug in the BLZ dongle and verify:

```
ls /dev/ttyUSB0
dmesg | grep -i "ch341\|cp210x\|ttyUSB"
```

You should see `ch341-uart converter now attached to ttyUSB0` (or similar for cp210x). If missing, your kernel lacks USB serial drivers – use the pre-built image or rebuild the kernel per Option B above.

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## Part 3 – Install Home Assistant (if not pre-installed)

The pre-built images already include Docker and HA. Skip this part if using a pre-built image.

```
# Install Docker
curl -fsSL https://get.docker.com | sudo sh
sudo usermod -aG docker $USER
# Log out and back in

# Avaota A1 needs --uts=host
mkdir -p ~/homeassistant/config
docker run -d \
  --name homeassistant \
  --restart=unless-stopped \
  --privileged \
  --network=host --uts=host \
  -e TZ=America/Los_Angeles \
  -v /run/dbus:/run/dbus:ro \
  -v ~/homeassistant/config:/config \
  --device=/dev/ttyUSB0 \
  ghcr.io/home-assistant/home-assistant:stable
```

Open `http://<BOARD_IP>:8123` after a few minutes.

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## Part A – Run ZHA with BLZ

Follow the **Docker** option in the ZHA BLZ Installation Guide:

```
HA_CONFIG=~/homeassistant/config
cd "$HA_CONFIG"
git clone --depth 1 \
  https://github.com/bouffalolab/haos_custom_zha_blz.git
mkdir -p custom_components
cp -r haos_custom_zha_blz/custom_components/zha \
  custom_components/
rm -rf haos_custom_zha_blz
docker restart homeassistant
```

In the HA web UI:

1. **Settings** > **Devices & Services** > **Add Integration**
2. Select **Zigbee Home Automation**
3. Choose **BLZ = Bouffalo Lab Zigbee radios: BL702/4/6**
4. Serial port: `/dev/ttyUSB0`, baud rate: `2000000`

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## Part B – Run Zigbee2MQTT with BLZ

Follow the **From Source** option in the Zigbee2MQTT BLZ Installation Guide:

```
# Prerequisites
sudo usermod -aG dialout $USER
sudo apt-get install -y mosquitto
sudo systemctl enable --now mosquitto

curl -fsSL https://deb.nodesource.com/setup_lts.x \
  | sudo -E bash -
sudo apt-get install -y nodejs git
sudo corepack enable
sudo corepack prepare pnpm@latest --activate

# Clone and build
git clone -b feat/blz \
  https://github.com/pine64/zigbee-herdsman.git
cd ~/zigbee-herdsman && pnpm install && pnpm run build

git clone -b feat/blz-local-dev \
  https://github.com/pine64/zigbee2mqtt.git
cd ~/zigbee2mqtt && pnpm install && pnpm run build
```

Create `~/zigbee2mqtt/data/configuration.yaml`:

```
homeassistant: true
mqtt:
  server: mqtt://127.0.0.1:1883
serial:
  port: /dev/ttyUSB0
  adapter: blz
  baudrate: 2000000
frontend:
  enabled: true
  port: 8099
```

Start: `cd ~/zigbee2mqtt && pnpm start`

In HA, add the **MQTT** integration (broker: `127.0.0.1`, port: `1883`). Devices appear via MQTT Discovery.

**Note:** Do not run ZHA and Z2M on the same dongle simultaneously. Stop one before starting the other.